that becomes a truth when he goes there in the tesseract (Chapter 29); and the laws of quantum gravity are a speculation until TARS extracts them from inside a black hole so for Cooper and Murph they become truth (Chapters 28 and 30).

For nineteenth-century physicists, Newton's inverse square law for gravity was an absolute truth. But around 1890 it was revolutionarily upended by a tiny observed anomaly in the orbit of Mercury around the Sun (Chapter 24). Newton's law is very nearly correct in our solar system, but not quite. This anomaly helped pave the way for Einstein's twentieth-century relativistic laws, which—in the realm of strong gravity—began as speculation, became an educated guess when observational data started rolling in, and by 1980, with ever-improving observations, evolved into truth (Chapter 4).

Revolutions that upend established scientific truth are exceedingly rare. But when they happen, they can have profound effects on science and technology.

Can you identify in your own life speculations that became educated guesses and then truth? Have you ever seen your established truths upended, with a resulting revolution in your life?

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5 In these realms, for example, the energy of light has huge quantum fluctuations. They are so huge that they warp space and time enormously and randomly. The fluctuating warpage is beyond the scope of Einstein's relativistic laws, and the warpage's influence on the light is beyond the scope of the light's quantum laws.

6 The phrase "fiery marriage" was coined by my mentor John Wheeler, who was superb at naming things. John also coined the words "black hole" and "wormhole" and the phrase "a black hole has no hair"; Chapters 14 and 5. He once described to me lying in a warm bath for hours on end, letting his mind soar in a search for just the right word or phrase.

7 Chapters 5, 6, and 8.